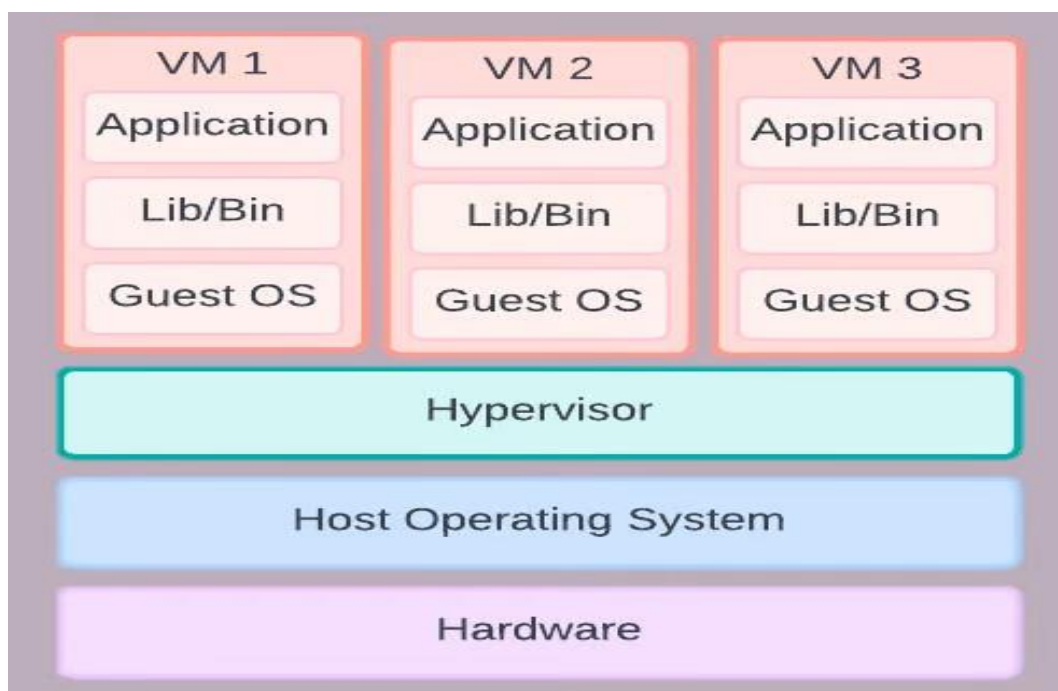


VIRTUAL MACHINE

A virtual machine (VM) is a software-based emulation of a physical computer, running on top of another physical computer. It allows users to run multiple operating systems and applications on a single physical server using virtualization technology. This enables cloud infrastructure to be elastic, efficient with resource utilization, and adaptable.

Architecture:



How do Virtual Machine Works?

Virtual machines use virtualization technology to create virtual hardware—or a virtual version of a computer on a physical machine. The physical machine on which the VMs run is called the **host**, and the VMs running on the host are called **guests**.

Each guest VM runs on an isolated partition on the host, completely separated from other guests. You can host multiple VMs on a single host machine, often a server, running on a software layer known as the **hypervisor**.

The hypervisor abstracts the host machine's physical resources, such as compute, memory, or storage, into a pool that can be provisioned and dynamically allocated to guest VMs as needed, providing more flexibility and increasing overall efficiency.

Types of VMs:

***.Process VM:** A process VM, also called an application virtual machine or managed runtime environment (MRE), creates a virtual environment of an OS while an app or single process is running and destroys it as soon as you exit. Process VMs enable creating a platform-independent environment that lets an app or process run the same way on any platform.

***.System VM:** A system VM (sometimes called hardware virtual machines) simulates a complete operating system, allowing multiple OS environments to live on the same machine. Typically, this is the type of VM people are referring to when they talk about "virtual machines." System VMs can run their own OS and applications, and a hypervisor monitors and distributes the physical host machine's resources between system VMs.

Benefits:

***.Scalability** - Cloud-based VMs make it easier to scale your applications, increasing availability and performance.

***.Portability** - A virtual machine is a single software package with hardware resources, an operating system, and all its applications.

***.Reduced footprint and costs**

***.Better security** - Virtual machines allow you to run multiple operating systems without impacting the host operating system.

***.Reliability** - Virtual machines and their components exist virtually and remain isolated from other guest VMs.

How are Virtual Machines used?

- **Consolidate servers:** Multiple physical machines can be reconfigured as a VM and run on a host alongside other VMs, allowing organizations to reduce sprawl.
- **Create development and test environments:** VMs can serve as isolated environments for testing and development that include full functionality but have no impact on the surrounding infrastructure.
- **Support DevOps:** VMs can easily be turned off or on, migrated, and adapted, providing maximum flexibility for development.
- **Enable workload migration:** The flexibility and portability that VMs provide are key to increasing the velocity of migration initiatives.
- **Improve disaster recovery and business continuity:** Replicating systems in cloud environments using VMs can provide an extra layer of security and certainty. Cloud environments can also be continuously updated.
- **Create a hybrid environment:** VMs provide the foundation for creating a cloud environment alongside an on-premises one, bringing flexibility without abandoning legacy systems.